

Weight Verification and Smart DISPATCH VALIDATION SYSTEM

Smart quality assurance solution
Industrial automation-based

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PROBLEM STATEMENT

- Missing components in packages
- Human errors during checking
- Slow manual validation process
- Weight mismatch issues



EXISTING SYSTEM

- Manual barcode verification
- Manual weight checking
- No automatic validation
- Slow processing speed
- High dependency on workers



PROPOSED SYSTEM

- Automated barcode scanning
- Real-time weight verification
- Automatic PASS/FAIL detection
- QR code label printing
- Industrial automation workflow
- User Interface



OBJECTIVES

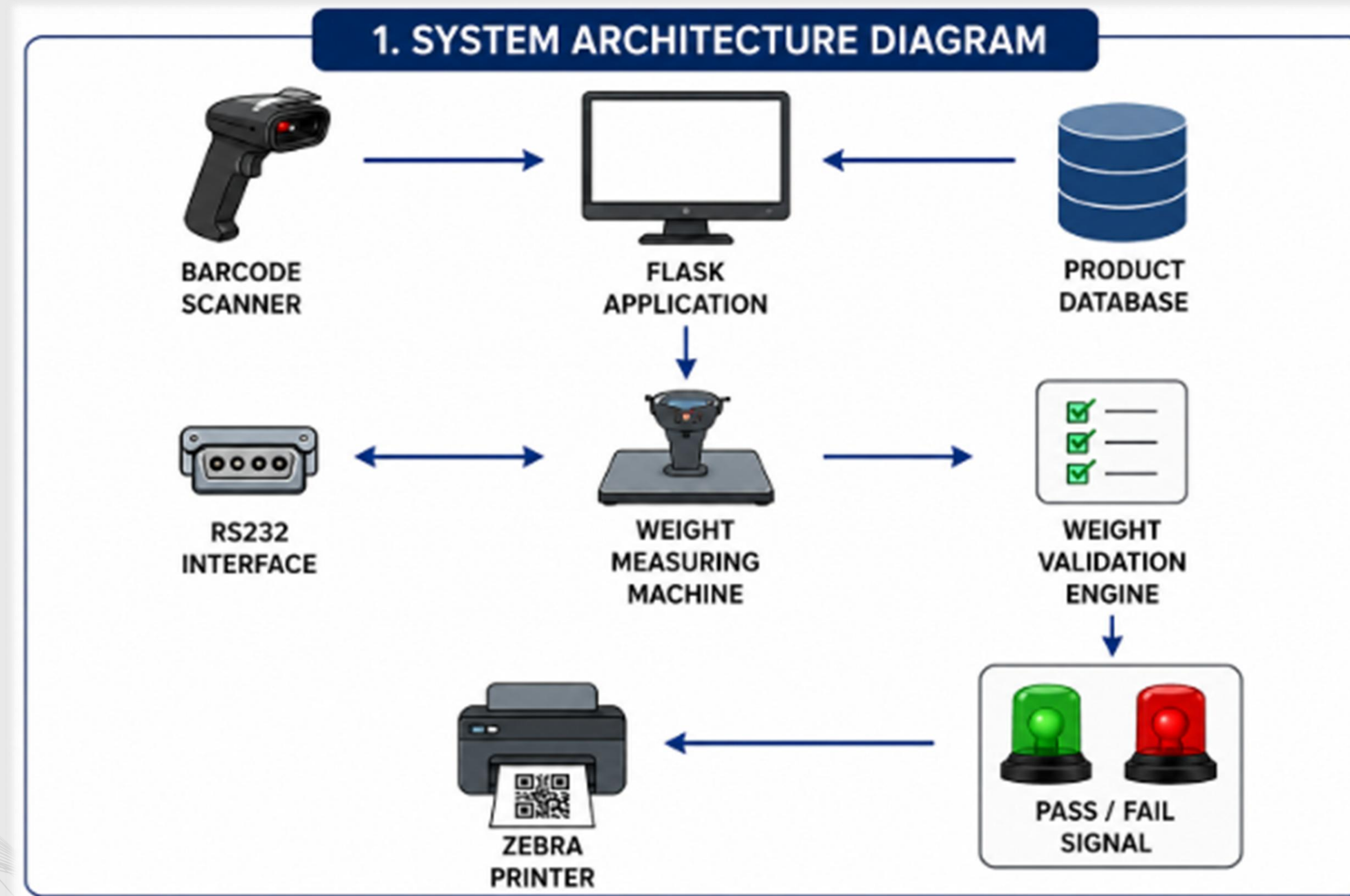
- Reduce manual errors
- Increase dispatch speed
- Improve quality assurance
- Enable smart automation
- Implement real-time verification



SYSTEM ARCHITECTURE

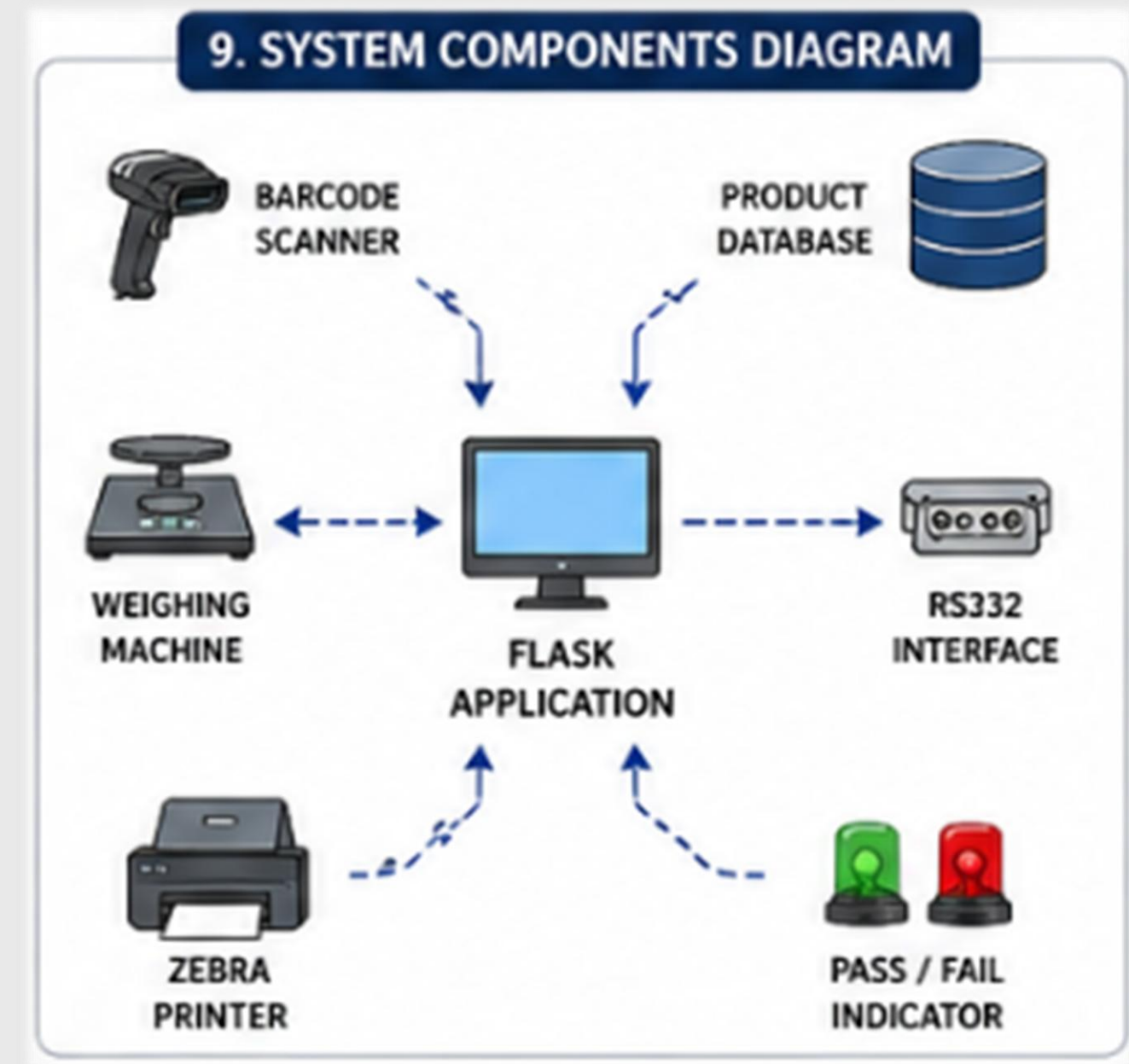
- Barcode scanner input
- RS232 communication
- Product database verification
- Weight validation engine
- Zebra QR printing system

BLOCK DIAGRAM

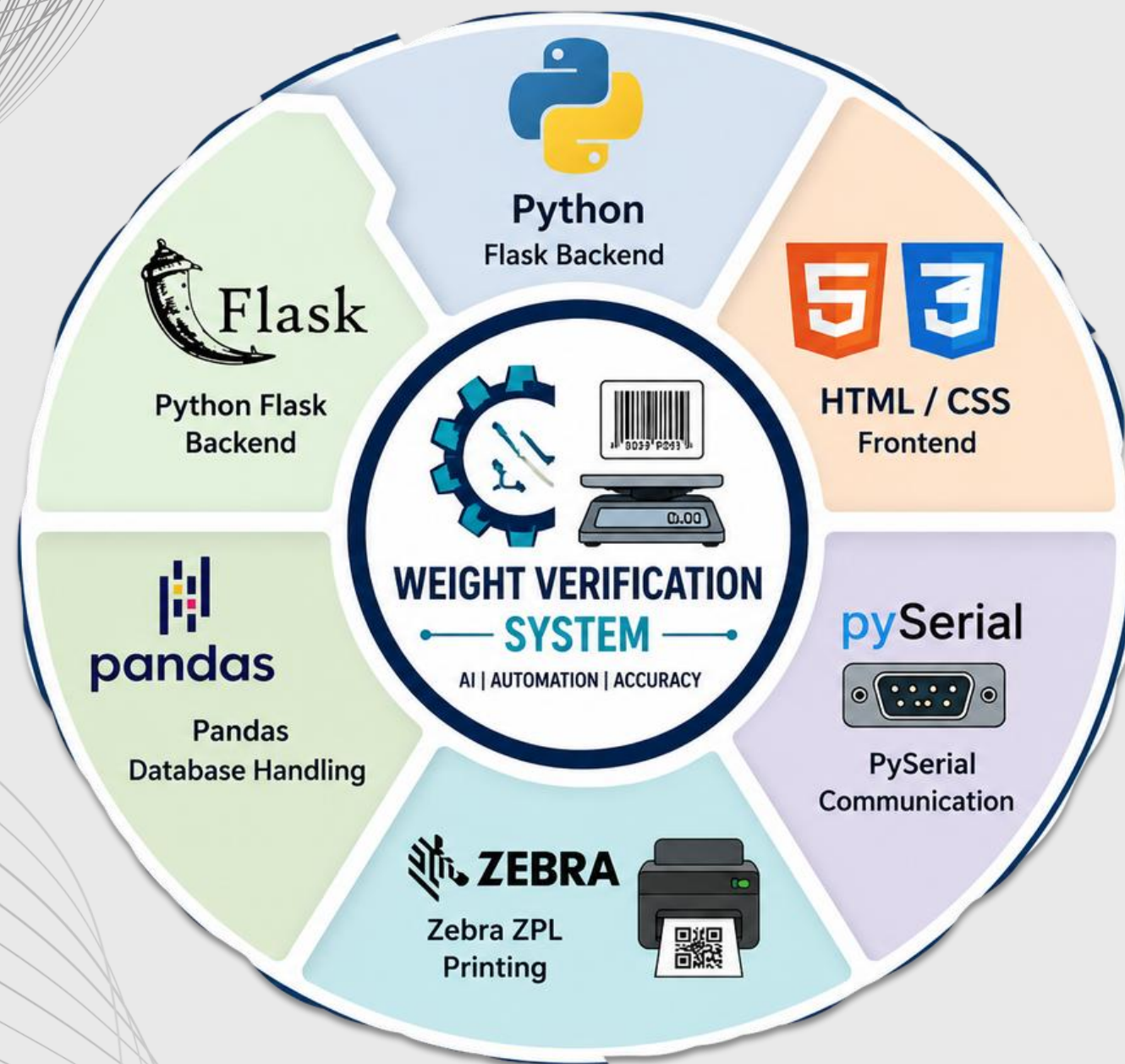


HARDWARE COMPONENTS

- Barcode Scanner
- Weighing Machine
- RS232 Converter
- Zebra Printer
- Computer System



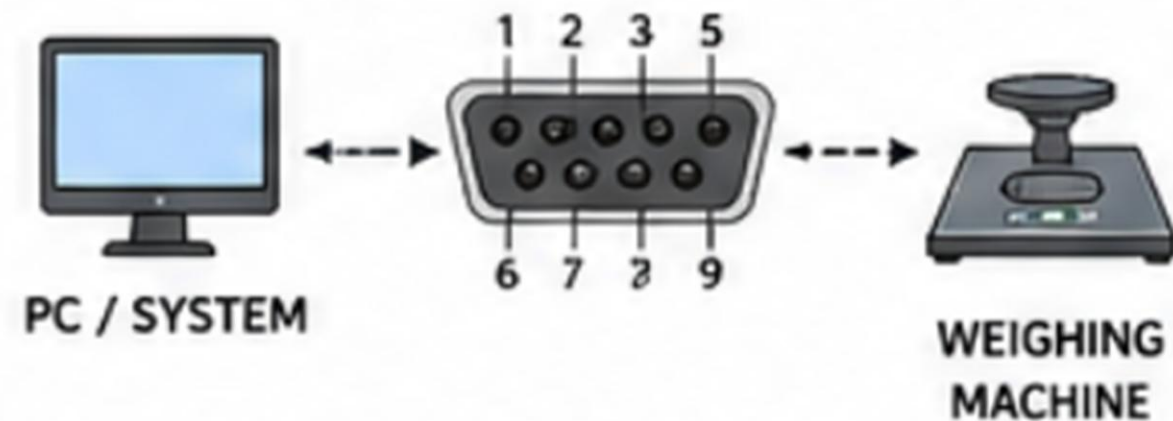
SOFTWARE & TECHNOLOGIES



- Python Flask backend
- HTML/CSS frontend
- Pandas database handling
- PySerial communication
- Zebra ZPL printing

RS232 COMMUNICATION

5. RS232 COMMUNICATION DIAGRAM



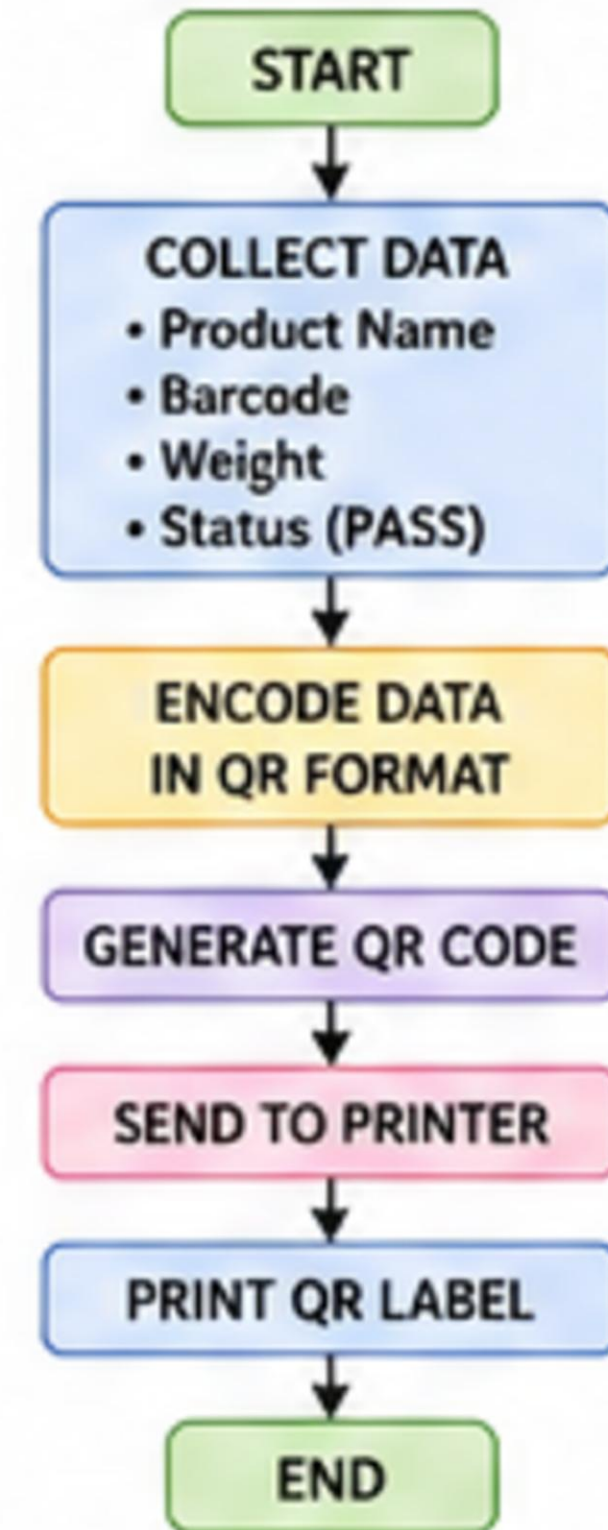
PIN NO.	PC (DB9)	WEIGHING MACHINE	SIGNAL
2	RXD	TXD	Receive Data
3	TXD	RXD	Transmit Data
5	GND	GND	Ground

- Transfers weight data
- Real-time serial communication
- Stable industrial protocol
- Machine-to-PC integration

QR PRINTING PROCESS

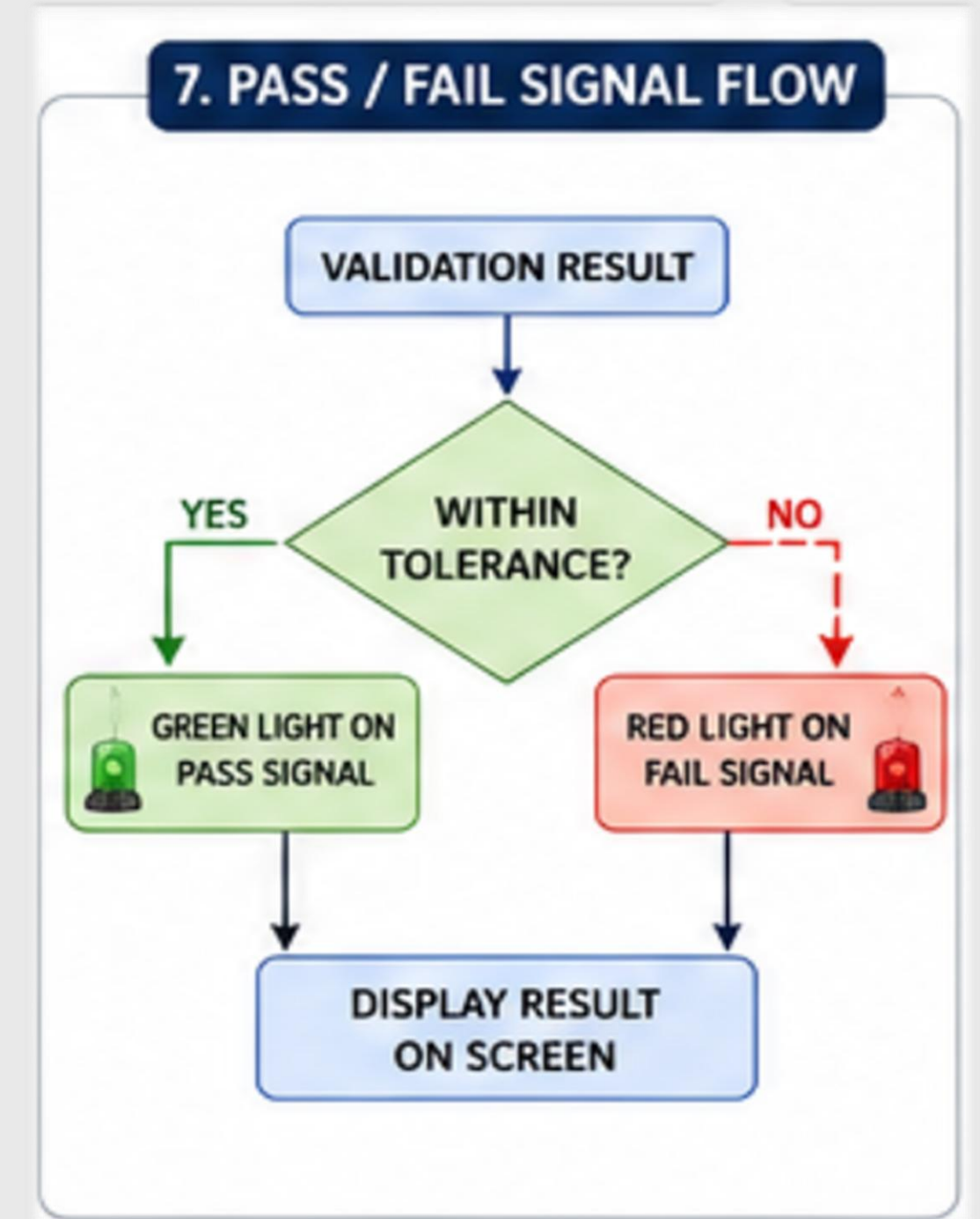
- QR generated after PASS
- Zebra printer integration
- QR contains product details
- Smart label generation

6. QR CODE GENERATION FLOW



WEIGHT VALIDATION LOGIC

- Expected weight retrieved
- Actual weight measured
- $\pm 5\%$ tolerance checking
- PASS/FAIL generation





FRONTEND DASHBOARD

- Industrial UI design
- Live weight display
- PASS/FAIL status
- Auto-focus barcode scanning
- Real-time validation screen

Weight Verification System

Industrial Product Validation Dashboard

● SYSTEM ONLINE

Barcode / Manual Input

Scan Barcode or Enter Number

VERIFY PRODUCT

Product Details

Product Name

Product B

Stored Weight

750 g

Live Weight

750g

Current Status

PASS

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [ ] [ ] ... [ ] [ ] X

(.venv-1) PS D:\project\AL_weight> python app.py
2026-05-22 09:39:11,248 - INFO - Using Printer: ZDesigner ZD421-300dpi ZPL
2026-05-22 09:39:11,300 - INFO - [x] QR Label Printed Successfully
2026-05-22 09:39:11,302 - INFO - 127.0.0.1 - - [22/May/2026 09:39:11] "POST /verify HTTP/1.1" 200 -
(.venv-1) PS D:\project\AL_weight> python app.py
2026-05-22 09:46:46,219 - WARNING - [!] Serial Port Not Connected (Demo Mode Enabled)
2026-05-22 09:46:46,219 - ERROR - could not open port 'COM3': OSError(22, 'The semaphore timeout period has expired.', None, 121)
2026-05-22 09:46:46,252 - INFO - [x] CSV DATA LOADED SUCCESSFULLY
* Serving Flask app 'app'
* Debug mode: off
2026-05-22 09:46:46,308 - INFO - WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.21.26.179:5000
2026-05-22 09:46:46,309 - INFO - Press CTRL+C to quit
2026-05-22 09:46:52,028 - INFO - 127.0.0.1 - - [22/May/2026 09:46:52] "GET / HTTP/1.1" 200 -
2026-05-22 09:46:52,245 - INFO - 127.0.0.1 - - [22/May/2026 09:46:52] "GET /static/style.css HTTP/1.1" 304 -
2026-05-22 09:46:55,651 - INFO - Scanned Barcode: 8901230000001
2026-05-22 09:46:55,651 - INFO - [!] DEMO MODE ENABLED
2026-05-22 09:46:55,652 - INFO - Expected Weight: 750.0
2026-05-22 09:46:55,652 - INFO - Current Weight: 750.0
2026-05-22 09:46:55,652 - INFO - [x] STATUS: PASS
2026-05-22 09:46:55,652 - INFO - Using Printer: ZDesigner ZD421-300dpi ZPL
2026-05-22 09:46:55,758 - INFO - [x] QR Label Printed Successfully
2026-05-22 09:46:55,759 - INFO - 127.0.0.1 - - [22/May/2026 09:46:55] "POST /verify HTTP/1.1" 200 -
2026-05-22 09:48:00,475 - INFO - Scanned Barcode: 4025515080930
2026-05-22 09:48:00,475 - INFO - [!] DEMO MODE ENABLED
2026-05-22 09:48:00,476 - INFO - Expected Weight: 1685.0
2026-05-22 09:48:00,476 - INFO - Current Weight: 1685.0
2026-05-22 09:48:00,476 - INFO - [x] STATUS: PASS
2026-05-22 09:48:00,476 - INFO - Using Printer: ZDesigner ZD421-300dpi ZPL
2026-05-22 09:48:00,614 - INFO - [x] QR Label Printed Successfully
2026-05-22 09:48:00,615 - INFO - 127.0.0.1 - - [22/May/2026 09:48:00] "POST /verify HTTP/1.1" 200 -
2026-05-22 09:49:51,401 - INFO - Scanned Barcode: 8901230000000
```

Google



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AI Mode

All

Exact matches

Visual matches

About this image

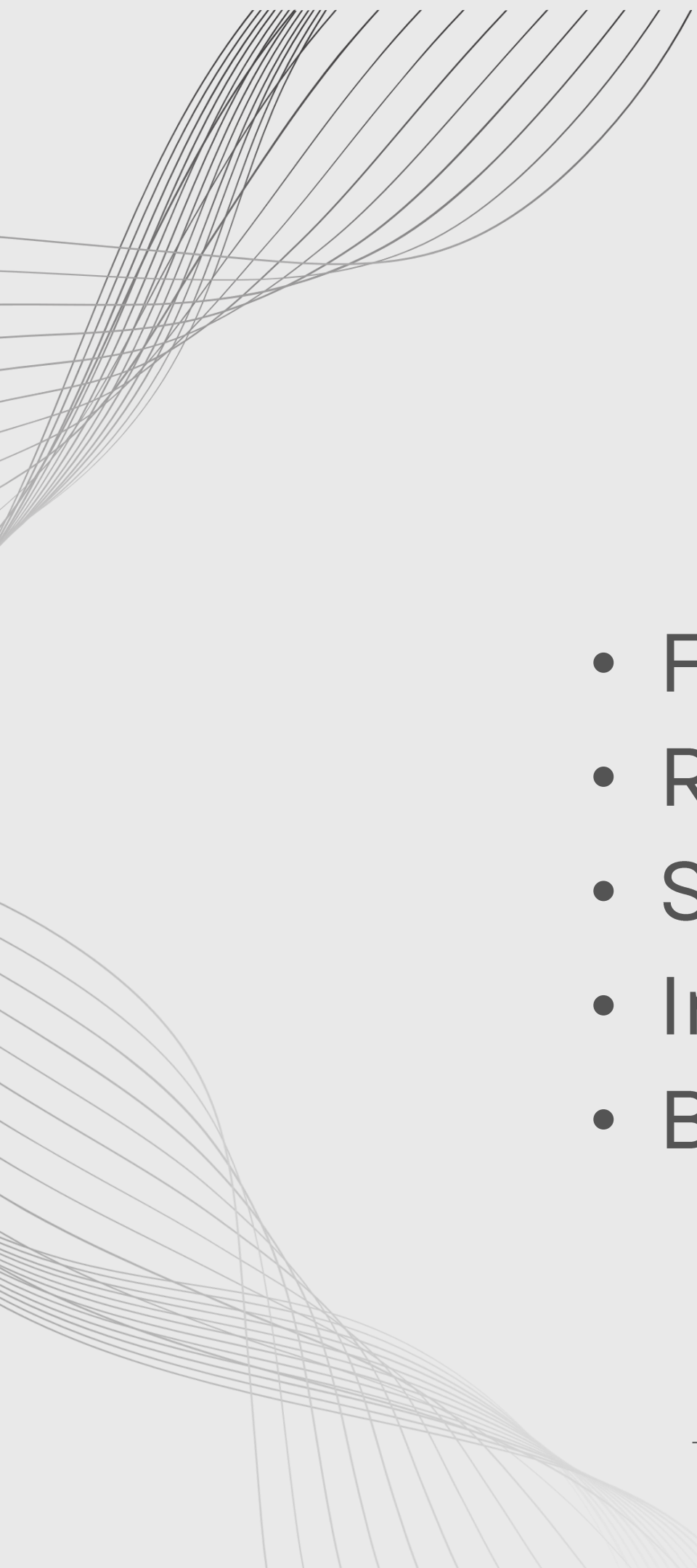
Feedback

RESULT: PASS PRODUCT: Product A BARCODE: 8901230000000 EXPECTED: 500.0
ACTUAL: 500.0 TIME: 2026-05-18 10:43:12.459887



RESULTS & OUTPUT

- Successful barcode scanning
- Accurate weight validation
- Automatic QR printing
- Faster dispatch processing
- Reduced human errors



ADVANTAGES

- Faster processing
- Reduced manual errors
- Smart automation
- Improved productivity
- Better quality control



FUTURE ENHANCEMENTS

- Cloud database integration
- IoT monitoring
- AI analytics dashboard
- ERP integration
- RFID support



CONCLUSION

- Successfully automated dispatch verification
- Improved industrial efficiency
- Reduced human errors
- Real-time weight validation achieved
- Smart QR printing implemented



THANK YOU